

Gravitational and Phenomenological Properties of Early Universe Defects

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Summary and Outlook

The Early Universe

Inflationary picture¹ → period of rapid expansion and cooling

Symmetry breaking and phase transitions take place

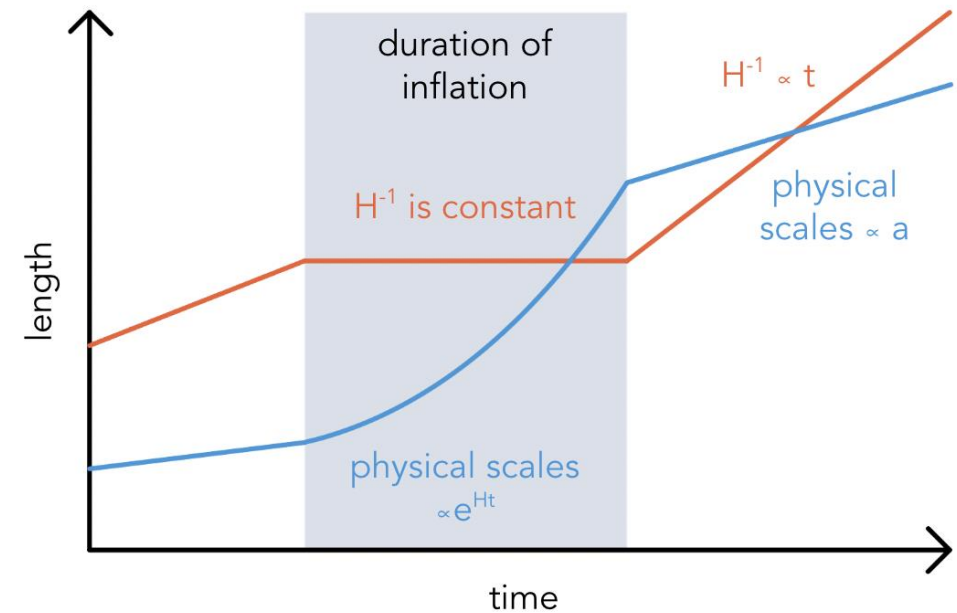
First order: proceeds via bubble nucleation

Second order: continuous change

Degenerate vacua → topological defects

Expect there to then be a period of reheating

Baryogenesis (by some mechanism)



¹ Guth, Phys. Rev. D 23 (1981) 347

QCD Axions and ALPs: The Strong CP Problem

Microscopic QCD Lagrangian (N_f flavours) admits a CP violating term

$$\mathcal{L} = -\frac{1}{4} G_a^{\mu\nu} G_{a\mu\nu} - \vartheta \frac{g_s^2}{32\pi^2} \underbrace{G_a^{\mu\nu}}_{\text{circled}} \tilde{G}_{a\mu\nu} + \bar{q} i \not{D} q - (\bar{q}_L \mathcal{M} q_R + \bar{q}_R \mathcal{M}^\dagger q_L)$$

Gluon field strength

$$G_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g_s f^{abc} A_\mu^b A_\nu^c$$
$$\tilde{G}_{\mu\nu} = \frac{1}{2} G_a^{\rho\sigma} \epsilon_{\mu\nu\rho\sigma}$$

Classically $\vartheta G_a^{\mu\nu} \tilde{G}_{a\mu\nu}$ is a total derivative

Non-perturbatively, QCD has non-trivial vacuum topology (instantons)

Invariant vacuum angle $\bar{\vartheta} = \vartheta + \arg \det \mathcal{M}$

Experimentally no CP violation observed, constraint: $|\bar{\vartheta}| \lesssim 10^{-10}$ –not explained by standard model

↳ “*Strong CP problem*,”²

² Kim & Carosi, Rev. Mod. Phys. 82 (2010) 557

QCD Axions and ALPs: The Peccei-Quinn Mechanism

Peccei-Quinn mechanism³: an elegant solution to the strong CP problem

Consider complex scalar field φ invariant under $U(1)_{PQ}$, $\varphi \rightarrow e^{i\epsilon} \varphi$, and a Lagrangian

$$\mathcal{L} = -\frac{1}{4} G_a^{\mu\nu} G_{a\mu\nu} - \frac{1}{2} |\partial_\mu \varphi|^2 - \frac{\lambda}{4} (|\varphi|^2 - \eta^2)^2$$

VEV of $\varphi = \eta e^{i\vartheta} = \eta e^{ia/\eta}$, $f_a \equiv \eta/N_{DW}$ spontaneously breaks global $U(1)_{PQ}$, gives dynamical Goldstone boson $a(x)$

The leading low energy effective interaction term is $\mathcal{L} \supset (\partial_\mu a) J_{PQ}^\mu$

$U(1)_{PQ}$ anomalous under QCD:

$$\partial_\mu J_{PQ}^\mu \sim \frac{g_s^2}{32\pi^2} G_a^{\mu\nu} \tilde{G}_{a\mu\nu}$$

Axion therefore couples as

$$\mathcal{L} \supset \frac{a}{f_a} \frac{g_s^2}{32\pi^2} G_a^{\mu\nu} \tilde{G}_{a\mu\nu}, \text{ such that } \vartheta_{eff} = \bar{\vartheta} + \frac{a}{f_a}$$

³ Peccei and Quinn, Phys. Rev. Lett. 38 (1977) 1440

QCD Axions and ALPs: Properties of Models

At QCD scale, strong coupling and non-perturbative effects activate ("QCD phase transition")

Instanton effects generate a vacuum potential for ϑ_{eff}

ϑ enjoys a shift symmetry $\vartheta = \vartheta + \frac{2\pi n}{N_{DW}}, n \in \mathbb{Z}$

Axion-like particle models (ALPs) are wider class with some other/general $U(1)$ symmetry⁴

- Allow wider range of values for m_a and f_a
- Alternative formation and dynamic evolution history

EFT Lagrangian:

$$\mathcal{L} = -\frac{1}{2}\eta^2\partial_\mu\vartheta\partial_\mu\vartheta - \frac{m_a^2 f_a^2}{N_{DW}^2} [1 - \cos(N_{DW}\vartheta)]$$

$\vartheta = \frac{a}{f_a}$

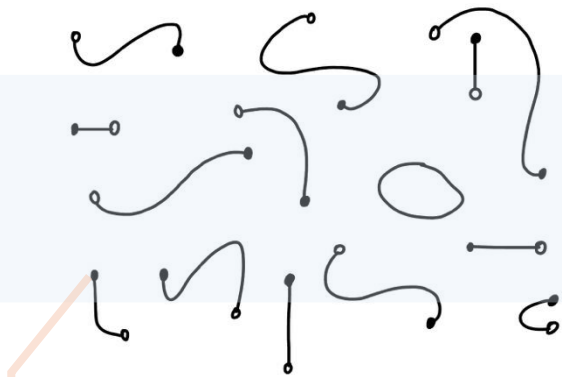
EM terms:

$$\mathcal{L}_{EM} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{\alpha N_{DW}\vartheta}{8\pi} F_{\mu\nu}\tilde{F}^{\mu\nu}$$

⁴ Ringwald, Phys. Dark Univ. 1 (2012) 116

Axions and Defects in Cosmology

String-domain wall networks are naturally formed from axion cosmology⁵

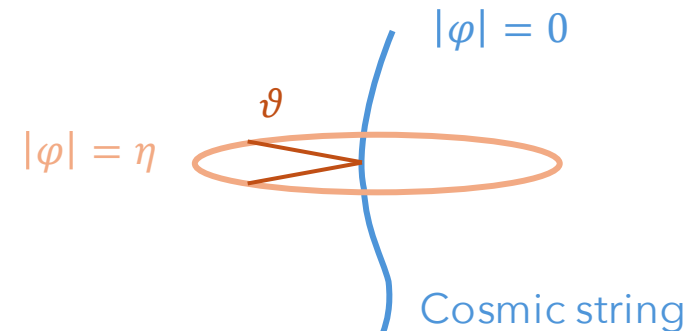


$U_{PQ}(1)$ broken after $T < \eta$

Field at symmetry breaking minimum

$$\langle \varphi \rangle = \eta e^{i\vartheta} = \eta e^{ia/\eta}, f_a \equiv \eta/N_{DW}$$

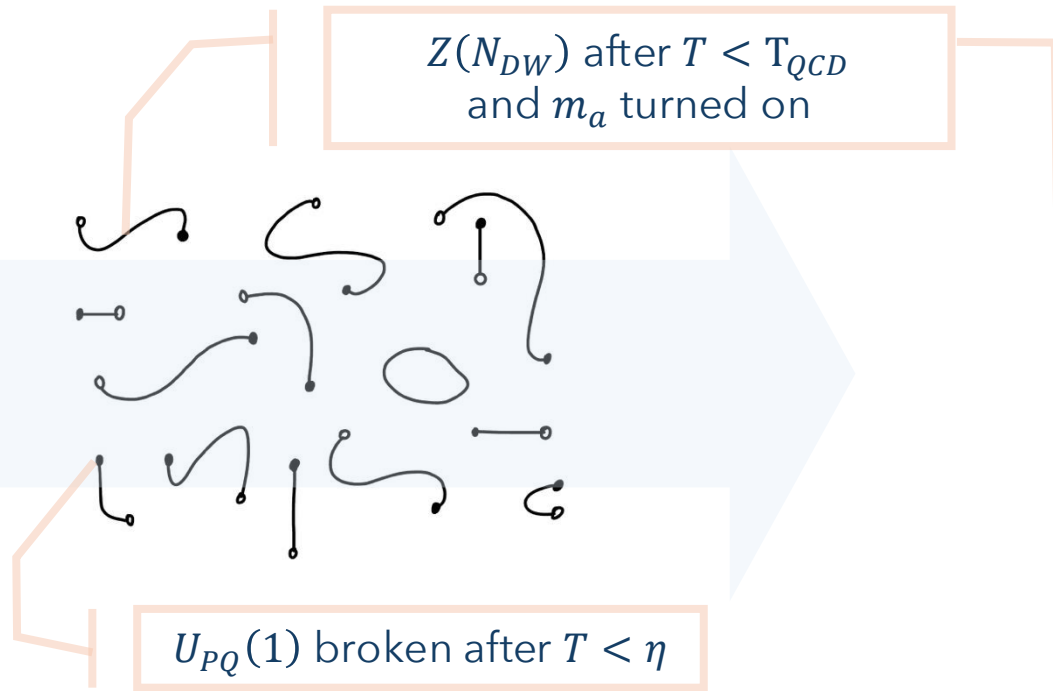
Vacuum manifold is S^1



⁵ Sikivie, Phys. Rev. Lett. 48 (1982) 1156

Axions and Defects in Cosmology

String-domain wall networks are naturally formed from axion cosmology⁵



$$V(\vartheta) = \frac{m_a^2 f_a^2}{N_{DW}^2} [1 - \cos(N_{DW}\vartheta)]$$

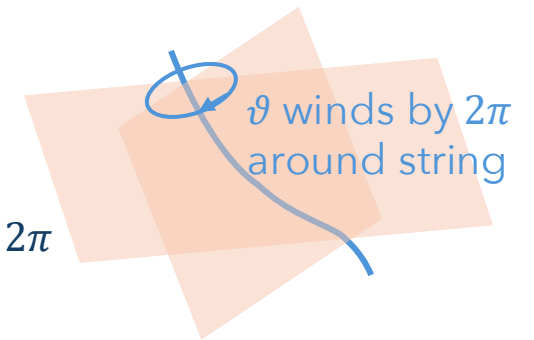
N_{DW} degenerate minima:

$$\vartheta = \frac{2\pi n}{N_{DW}}, \quad n = 0, \dots, N_{DW} - 1$$

e.g. $N_{DW} = 4$:

Walls separating

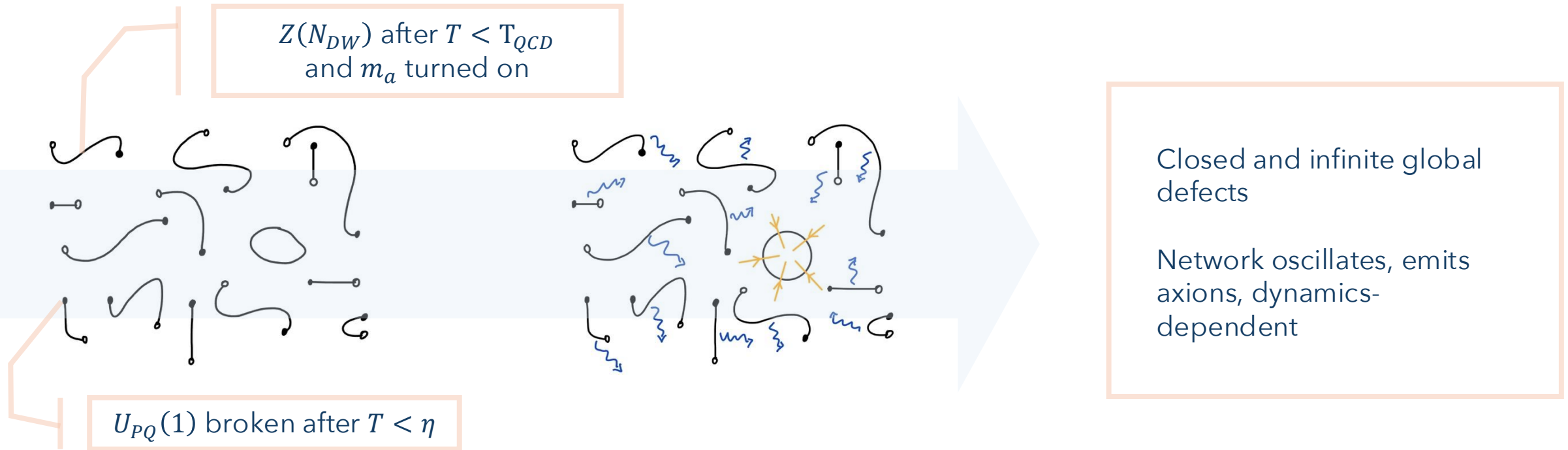
$$\vartheta = 0 \rightarrow \frac{\pi}{2} \rightarrow \pi \rightarrow \frac{3\pi}{2} \rightarrow 2\pi$$



⁵ Sikivie, Phys. Rev. Lett. 48 (1982) 1156

Axions and Defects in Cosmology

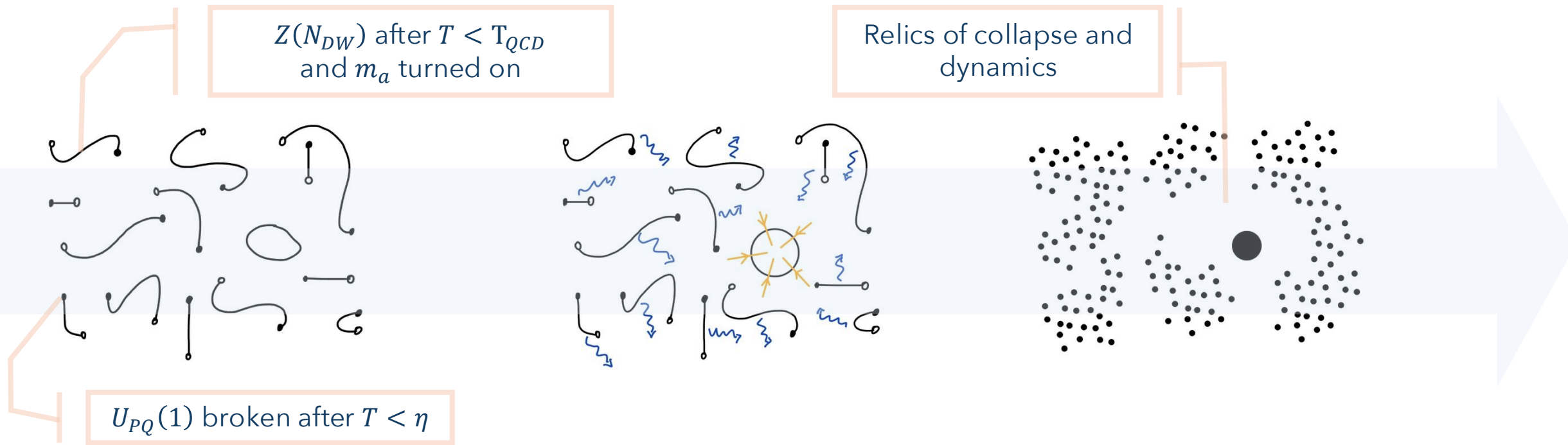
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Axions and Defects in Cosmology

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Axions and Defects in Cosmology

Formation determined by symmetry breaking scale

Pre-inflation

defects trivially diluted by inflation

Post-inflation

$N_{DW} \neq 1 \rightarrow$ domain wall problem

$N_{DW} = 1 \rightarrow$ inherently unstable
usual case for QCD axion models

During inflation

possible in ALP models
large defects possible

Isocurvature perturbations⁶

- PQ broken before inflation

$$\delta a \sim \frac{H_I}{2\pi}$$

- Generates "misalignment angle"
- After QCD phase transition, translates to modified dark matter distribution
 - Temperature, polarisation of CMB
- Observations constrain misalignment angle and so inflationary Hubble scale

⁶ Turner & Wilczek, Phys. Rev. Lett. 66 (1991) 5

Axions and Defects in Cosmology

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Standardly, $10^9 \text{GeV} \leq f_a \leq 10^{12} \text{GeV}$, string-wall network forms post inflation

$N_{DW} \neq 1 \rightarrow$ **domain wall problem**⁵

- Network is stable
- Domain wall energy density falls as a^{-1}
- Solution: bias potential

$N_{DW} = 1 \rightarrow$ **unstable**

- Network naturally collapses
- e.g. Lazarides-Shafi mechanism⁷

⁵ Sikivie, Phys. Rev. Lett. 48 (1982) 1156

⁷ Lazarides & Shafi, Phys. Lett. B 115 (1982) 21

Axions and Defects in Cosmology

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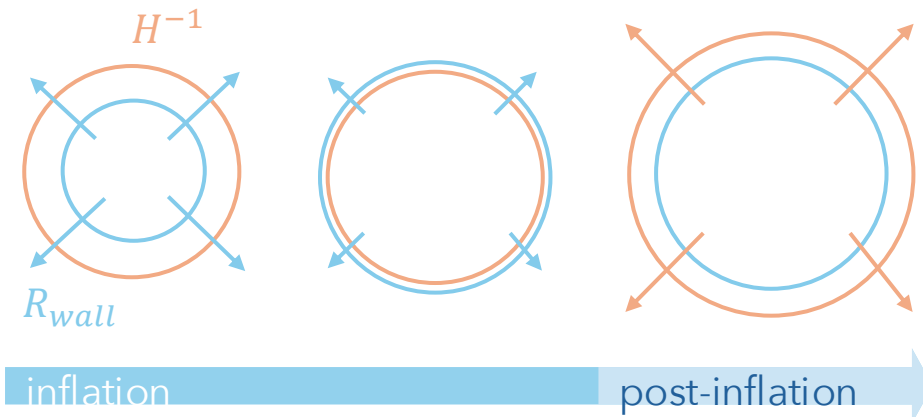
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usual case for QCD axion models

During inflation

possible in ALP models
large defects possible

ALP f_a and m_a less constrained

- Symmetry breaking during inflation⁸
- Defect network become super-horizon
- Re-enter post-inflation



⁸ Rubin, Khlopov & Sakharov, JCAP 09 (2001) 001; Ge, Guo & Liu, arXiv:2309.01739 (2023)

Final States of Defects

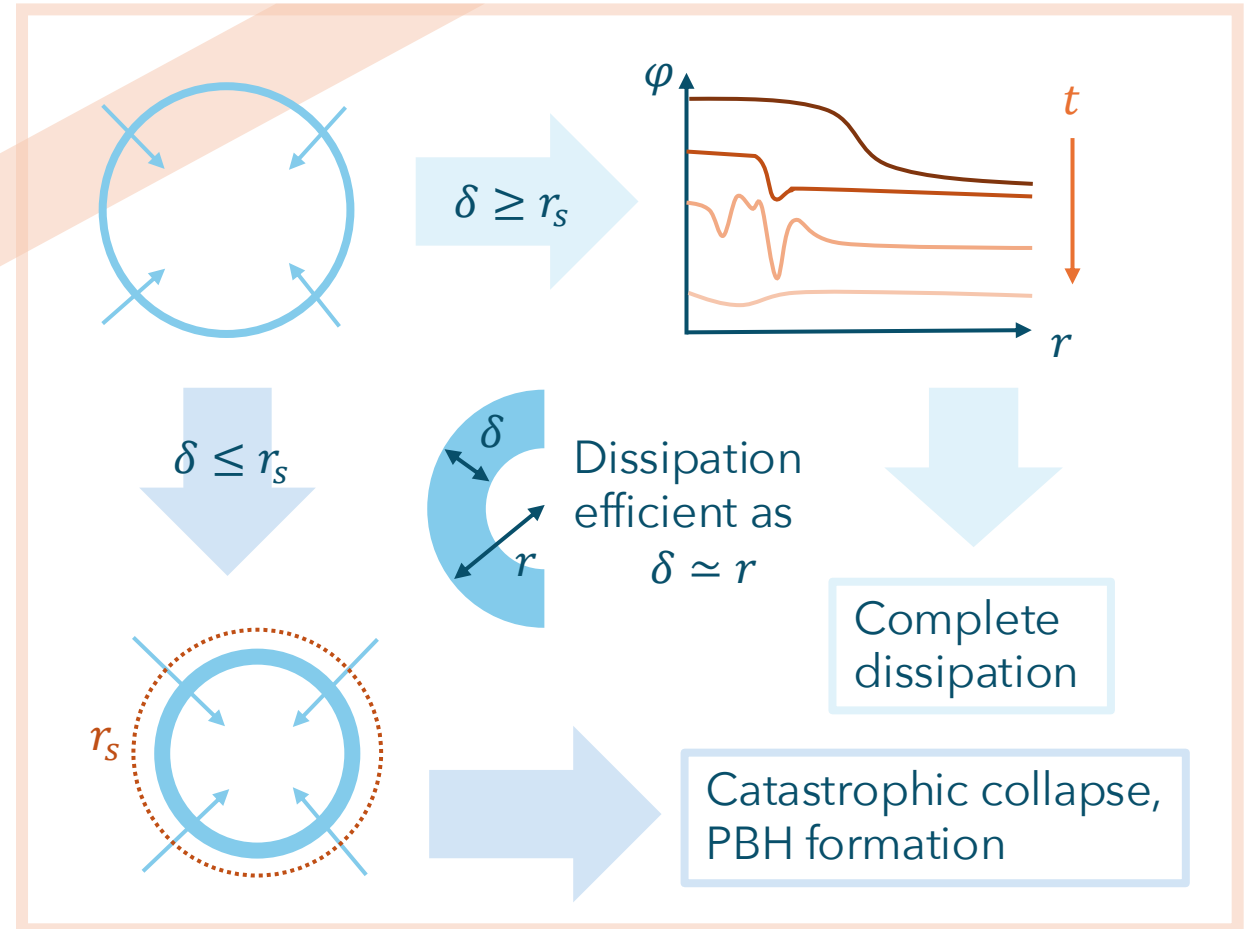
Final products depend on evolution pathway

Primordial black holes (PBH) can result from

- Closed wall collapse⁹
- Overdensities
- Closed string loop collapse

Emissions

- From defect/defect network collapse
- As Hawking radiation from PBH



⁹ Vilenkin, Phys. Rev. D 23 (1981) 852; Ipser & Sikivie, Phys. Rev. D 30 (1984) 712

Monopole Bags in Flat Space

't Hooft-Polyakov monopole¹⁰ formed from symmetry breaking in Yang-Mills theory e.g. GUT

Consists of core region of size $r_{\text{core}} \sim 1/\alpha M_m$ obeying original symmetries

Can be described as infinitesimal domain wall between interior (de Sitter) and exterior (asymptotically flat)

↳ obeying Israel-Darmois conditions

↳ topologically stable

↳ non-singular solution

Those with electrical excitations: **dyons**

$$\mathcal{L} \supset -\frac{1}{2} D_\mu \Phi^a D^\mu \Phi^a - \frac{\lambda}{4} (\Phi^a \Phi^a - \xi^2)^2 - \frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu}$$

$$ds^2 = -(1 - H^2 r^2) dt^2 + (1 - H^2 r^2)^{-1} dr^2 + r^2 d\Omega^2$$

$$H^2 = \frac{8\pi}{3} V(0) = \frac{2\pi\lambda\xi^4}{3}$$

$$h_{\mu\nu}^+ - h_{\mu\nu}^- = 0$$

$$K_{\mu\nu}^+ - K_{\mu\nu}^- = \left(-S_{\mu\nu} + \frac{1}{2} h_{\mu\nu} S \right)$$

¹⁰ 't Hooft, Nucl. Phys. B 79 (1974) 276; Polyakov, JETP Lett. 20 (1974) 194

Monopole Bags in Flat Space

"Witten effect"¹¹ describes dyon formation by placing monopole in an axionic background

$$\mathcal{L} \supset -\frac{1}{2} D_\mu \Phi^a D^\mu \Phi^a - \frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \frac{\alpha N_{DW} \vartheta}{8\pi} F_{\mu\nu}^a \tilde{F}^{a\mu\nu}$$



Modified Maxwell equation by
conserved "Witten current" ↗

$$\partial_\nu \left(F^{\mu\nu} - \frac{\alpha}{2\pi} \vartheta \tilde{F}^{\mu\nu} \right) = j_E^\mu,$$

$$\partial_\nu \tilde{F}^{\mu\nu} = j_M^\mu$$



$$\vec{\nabla} \cdot \vec{E} - \vec{\nabla} \cdot \left(\frac{\alpha}{2\pi} \vartheta \vec{B} \right) = \rho_E$$

¹¹ Witten, Phys. Lett. B 86 (1979) 283

Monopole Bags in Flat Space

Electrically and magnetically charged objects obey the DSZ quantisation condition¹²

$$q_1 q_{m2} - q_2 q_{m1} = 2\pi n$$

Taking $n = 1$ and minimum electric charge of 't Hooft-Polyakov monopole $q = e/2$, minimum magnetic charge is

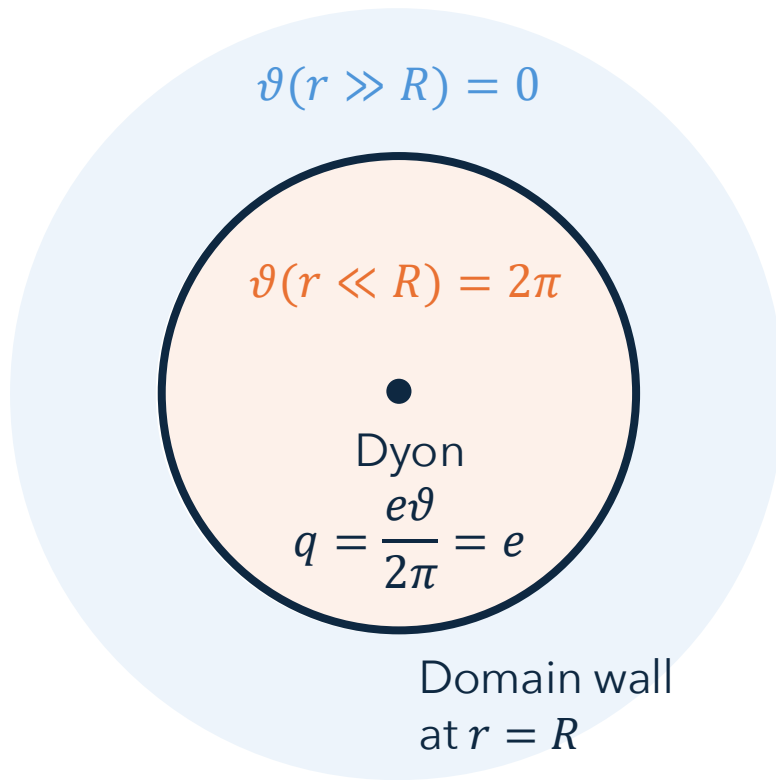
$$q_m = \frac{4\pi g}{e}$$

For which the electric charge is determined by background value ϑ :

$$q = ne + \frac{e\vartheta g}{2\pi} \quad \leftarrow \text{taking } g = 1 \text{ henceforth}$$

¹² Dirac, Proc. Roy. Soc. A 133 (1931) 60; Schwinger, Phys. Rev. 144 (1966) 1087; Zwanziger, Phys. Rev. 176 (1968) 1489

Monopole Bag in Flat Space



$$B_r = \frac{q_m}{4\pi} \frac{1}{r^2} = \frac{1}{er^2}$$

$$E_r = -\frac{e^2}{8\pi^2} \vartheta(r) B_r(r) + \frac{e}{r^2} = \frac{e}{4\pi r^2} \left(1 - \frac{\vartheta(r)}{2\pi}\right)$$

Central dyon electric field entirely shielded

EM energy of system goes as $\vec{E}^2 + \vec{B}^2$

↳ effective “repulsion” acting on wall

← Composite defect = “**monopole bag**”¹³

Monopole Bag in Flat Space

$$\mathcal{L} = -\frac{1}{2}\eta^2\partial_\mu\vartheta\partial_\mu\vartheta - \frac{m_a^2 f_a^2}{N_{DW}^2} [1 - \cos(N_{DW}\vartheta)] - \frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \frac{\alpha N_{DW}\vartheta}{8\pi} F_{\mu\nu}^a \tilde{F}^{a\mu\nu}$$

The equation of motion of ϑ is

$$0 = f_a^2 \ddot{\vartheta} - f_a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \vartheta}{\partial r} \right) + m_a^2 f_a^2 \sin \vartheta - \frac{e^2}{32\pi^3 r^4} \left(1 - \frac{\vartheta}{2\pi} \right)$$

Energy for the stationary axion configuration is then

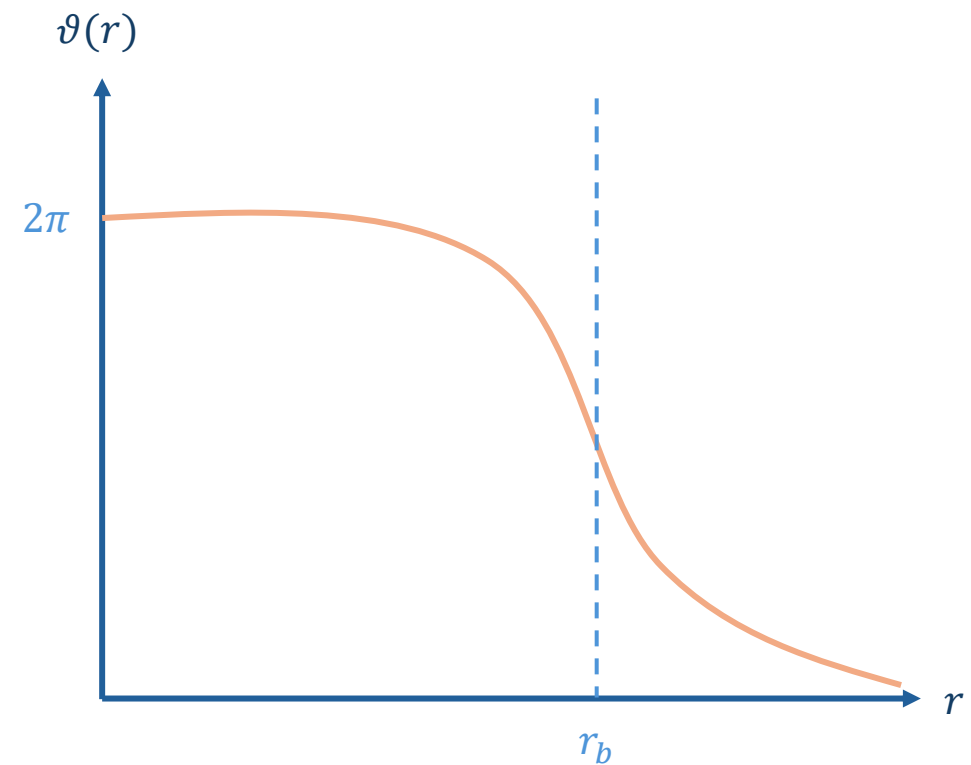
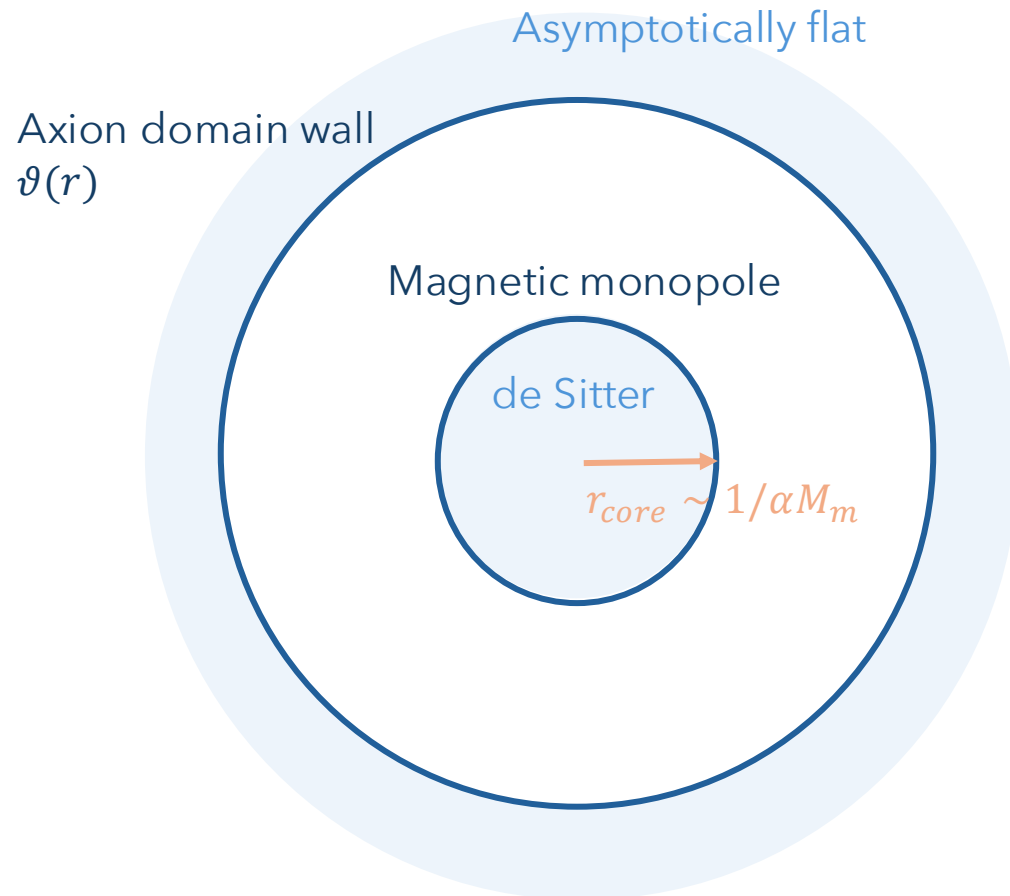
$$E = \int dr r^2 \left[\frac{1}{2} (\vec{\nabla} \vartheta)^2 + \boxed{m_a^2 f_a^2 (1 - \cos \vartheta) + \frac{e^2}{32\pi^2 r^4} \left(1 - \frac{\vartheta}{2\pi} \right)^2} \right] \quad \text{with } V(\vartheta) \text{ indicated by an arrow}$$

Solving the equation of motion for minimised effective potential $V(\vartheta)$

$$\vartheta(r) = 2\pi \left[1 - \exp\left(\frac{r_b}{r}\right) \right], \quad r_b = \frac{e^2}{8\pi^2 f_a}$$

Gravitating Monopole Bags

YK & Takayama, 2604.26191



Gravitating Monopole Bags

YK & Takayama, 2604.26191

Applying the metric ansatz $ds^2 = -N(r)C(r)dt^2 + C(r)^{-1}dr^2 + r^2d\Omega^2$

Minimally coupled Lagrangian

$$\mathcal{L} = \frac{M_{Pl}^2}{2} \mathcal{R} - \frac{1}{2} \partial_\mu a \partial_\mu a - m_a^2 f_a^2 \left(1 - \cos\left(\frac{a}{f_a}\right) \right) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{\alpha a}{8\pi f_a} F_{\mu\nu} \tilde{F}^{\mu\nu}$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, $A_\mu = (A_0(r), 0, 0, -q_m \cos(\theta))$

Finding equations of motion for A_0 , a and $F_{\mu\nu}$

$$A'_0 = \sqrt{N(r)} \frac{2\pi q f_a - a(r) \alpha q_m}{2\pi f_a r^2}$$

$$\square a - m_a^2 f_a \sin\left(\frac{a}{f_a}\right) + \frac{\alpha}{8\pi f_a} F_{\mu\nu} \tilde{F}^{\mu\nu} = 0$$

$$\nabla_\nu \left(-\frac{1}{4} F^{\mu\nu} + \frac{\alpha a}{8\pi f_a} \tilde{F}^{\mu\nu} \right) = 0$$

Taking the difference between (t, t) and (r, r) components of Einstein equations:

$$\rightarrow a' = \frac{N'}{8\pi N r}$$

Gravitating Monopole Bags

YK & Takayama, 2604.26191

Energy-momentum tensor turns out to be

$$T_{\nu}^{\mu} = \left(-\frac{1}{2} \partial^{\beta} a \partial_{\beta} a - m_a^2 f_a^2 \left(1 - \cos \left(\frac{a}{f_a} \right) \right) - \frac{1}{4} F_{\beta\gamma} F^{\beta\gamma} \right) \delta_{\nu}^{\mu} + \partial^{\mu} a \partial_{\nu} a - F^{\mu\beta} F_{\beta\nu}$$

We can constrain the relation between a and C using

equation of motion for C



$$-4\pi^2 f_a^3 r^4 \sin \left(\frac{a}{f_a} \right) + 4\pi^2 f_a^2 r^4 C a'' + 16\pi^3 f_a^2 r^5 a'^3 C + 4\pi^2 f_a^2 r^3 (rC' + 2C) a' - \alpha^2 q_m^2 a + 2\alpha q q_m \pi f_a = 0$$

$$-\frac{1}{2} a'^2 C - m_a^2 f_a^2 \left(1 - \cos \left(\frac{a}{f_a} \right) \right) - \frac{q_m^2}{2r^4} - \frac{A_0'^2}{2N} = \frac{rC' + C - 1}{8\pi r^4}$$

↪ (t, t) component of Einstein equations

Solve equations numerically using known boundary conditions

$$a(r_{core}) = 2\pi f_a, a(r \rightarrow \infty) = 0$$

$$C(r_{core}) = 1, C(r \rightarrow \infty) = 1$$

$$N(r_{core}) = N_0, N(r \rightarrow \infty) = 1$$

Gravitating Monopole Bags

YK & Takayama, 2604.26191

Total mass can be determined from integrating the energy density

$$T_t^t \equiv \rho = -\frac{1}{2}a'^2C + m_a^2f_a^2 \left(1 - \cos\left(\frac{a}{f_a}\right)\right) + \frac{(2\pi f_a q - \alpha a q_m)^2 + 4\pi^2 f_a^2 q_m^2}{8\pi^2 f_a^2 r^4}$$

We can constrain the relation between a and C using

equation of motion for C



$$-4\pi^2 f_a^3 r^4 \sin\left(\frac{a}{f_a}\right) + 4\pi^2 f_a^2 r^4 C a'' + 16\pi^3 f_a^2 r^5 a'^3 C + 4\pi^2 f_a^2 r^3 (rC' + 2C)a' - \alpha^2 q_m^2 a + 2\alpha q q_m \pi f_a = 0$$

$$-\frac{1}{2}a'^2C - m_a^2f_a^2 \left(1 - \cos\left(\frac{a}{f_a}\right)\right) - \frac{q_m^2}{2r^4} - \frac{A_0'^2}{2N} = \frac{rC' + C - 1}{8\pi r^4}$$

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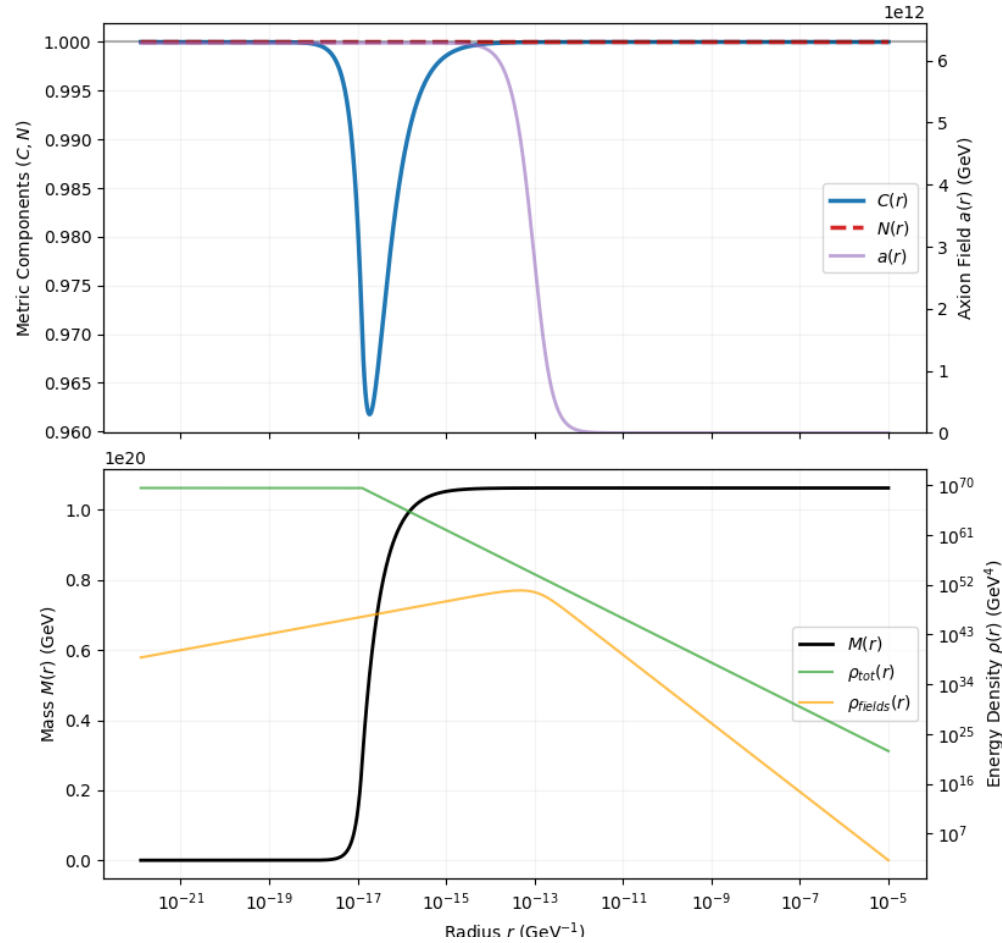
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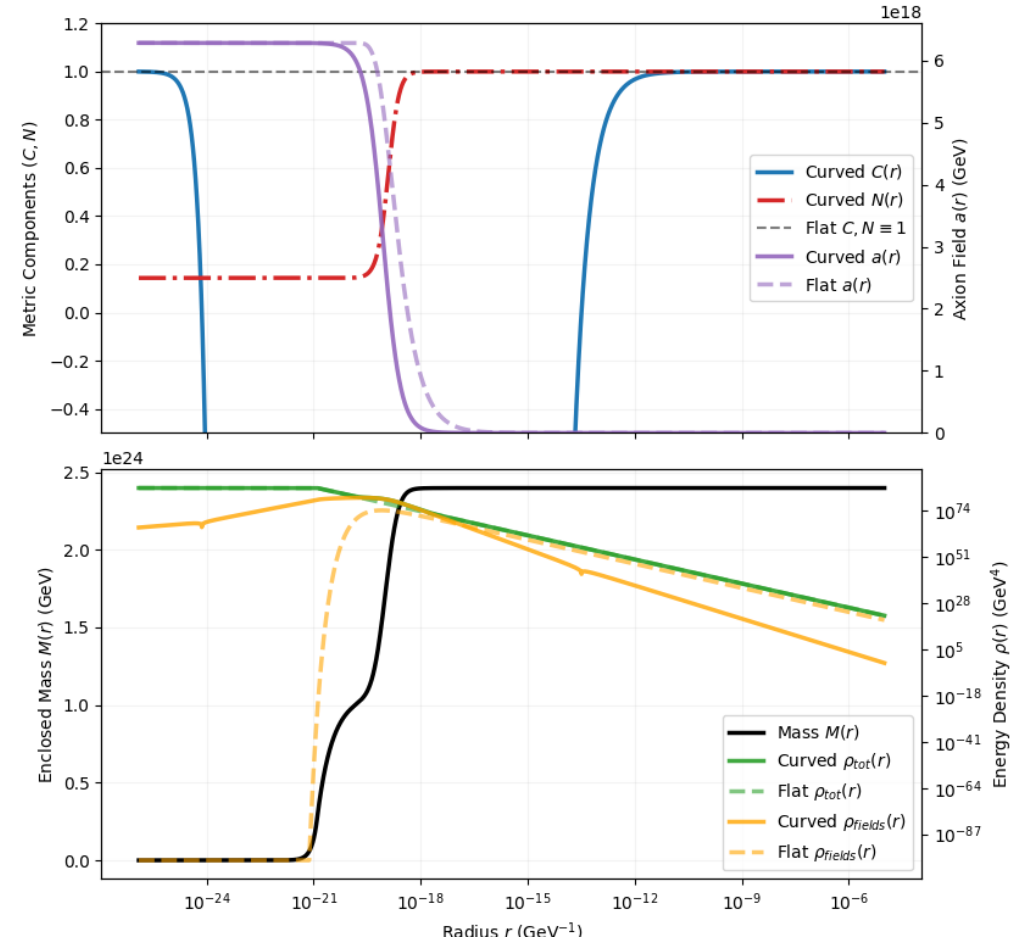
Gravitating Monopole Bags

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$$q = 1, q_m = 4\pi, \alpha = 1/4\pi$$

$$f_a = 10^{12} \text{ GeV}, m_a = 10^{-15} \text{ GeV}, M_m = 10^{18} \text{ GeV}$$

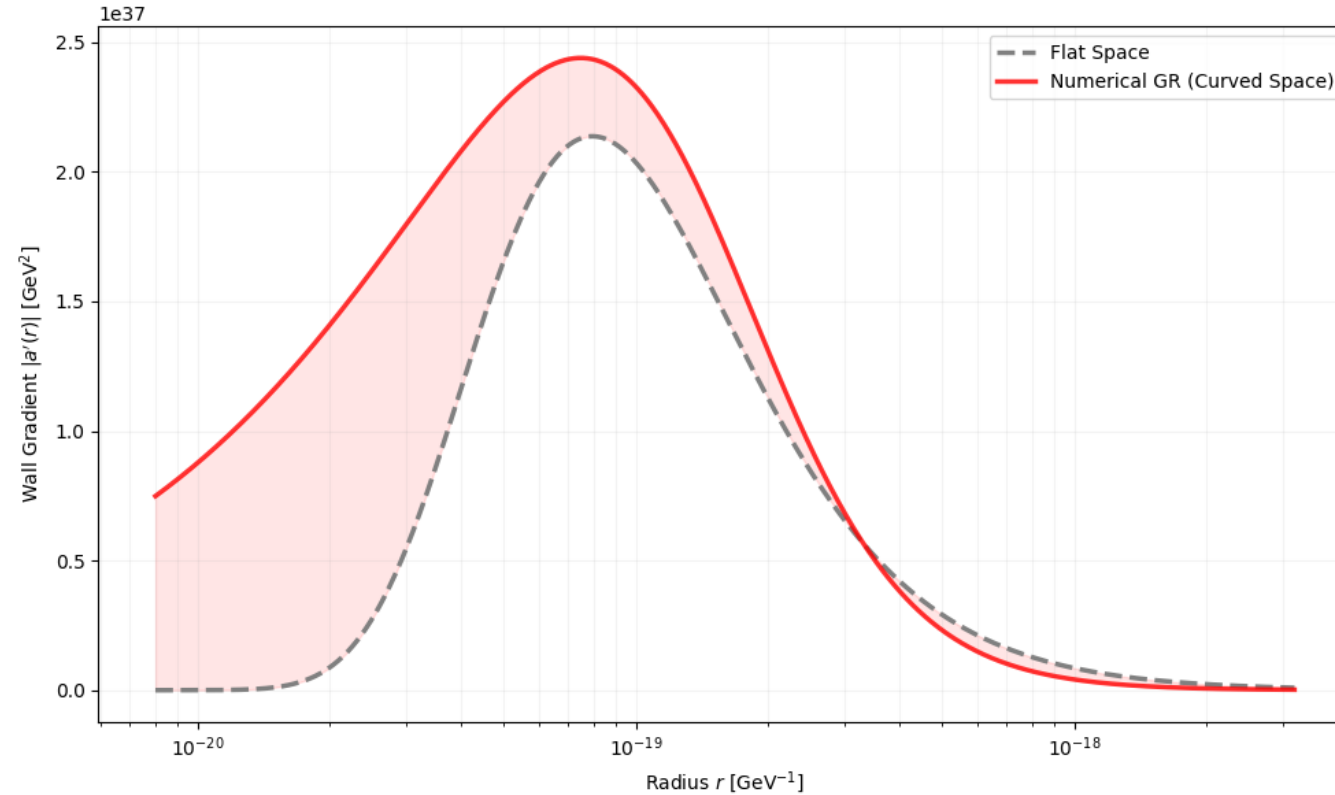


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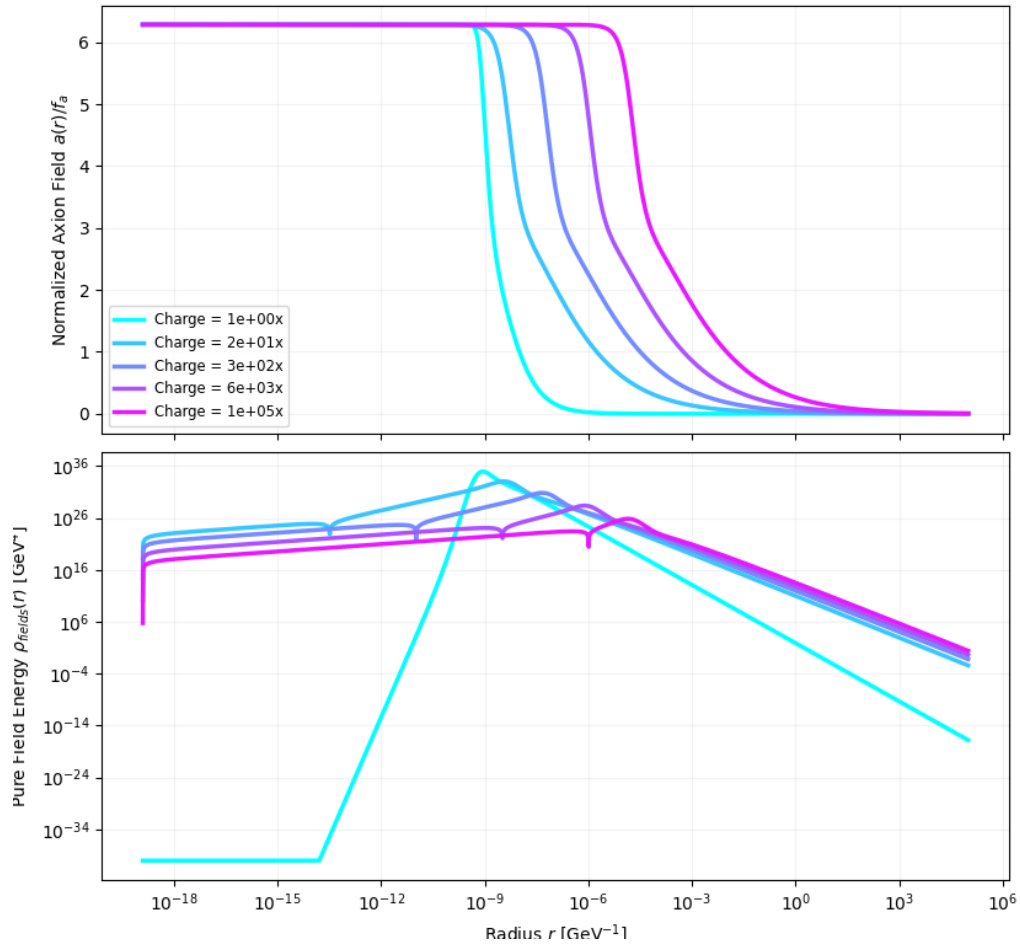
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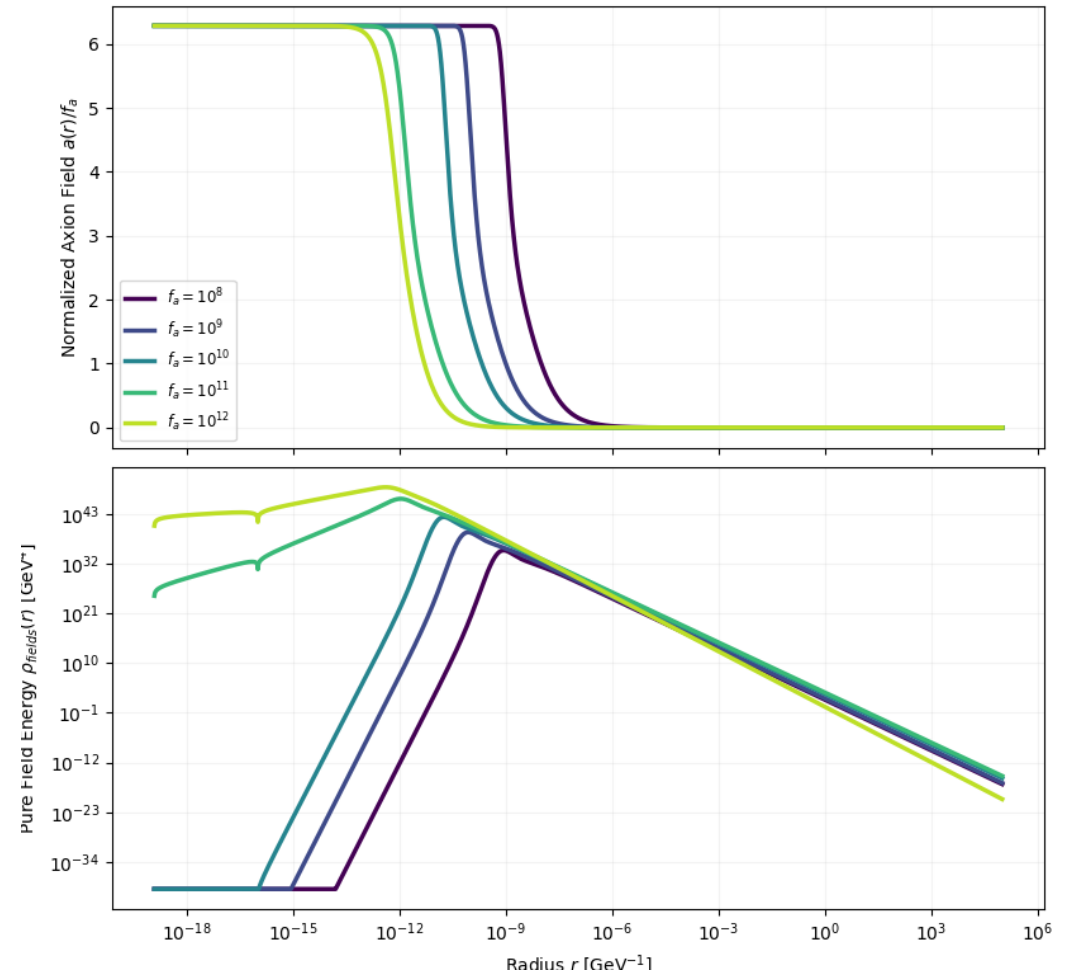
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$$q = 1, q_m = 4\pi, \alpha = 1/4\pi, M_m = 10^{18} \text{ GeV}$$

Physical and Theoretical Implications

Alternative evolution pathway to standard PBH formation from defects (example: ALP domain walls)

- topological structure at the centre is determined by the associated Witten effect
- variation in the value of $a(r \rightarrow 0)$ away from $2\pi f_a$ incurs a great cost in EM energy density
- In place of trivialisation of the external axionic profile, EM and gravitational influences expected to balance
- Taking $R = 10^{15} \text{GeV}^{-1}$ and $\sigma \sim m_a f_a^2$, minimum formation condition is

$$m_a f_a \gtrsim \frac{M_{Pl}}{R} \approx 104 (\text{GeV}^2)$$

↳ expect $m_a = 10^{-20} \text{GeV}$, $f_a = 10^{18} \text{GeV}$ to not be able to reliably form PBHs.

Physical and Theoretical Implications

Projected consequences of deformed profile

- Asphericity¹⁴ and angular momentum¹⁵ affect PBH formation
 - Deformed profile $\Rightarrow$ modified energy distribution, thick domain wall
 - Contributions of aspherical perturbations $\Rightarrow$ emissions
 - ↳ Gravitational waves
 - ↳ EM effects
 - ↳ Backreaction to medium
- 
- Shear and frictional components
 - Perturbations inducing rotation/asphericity
 - Interactions with primordial magnetic fields
 - Gravitational, optical lensing effects

¹⁴ Yoo, Harada & Okawa, Phys. Rev. D 102 (2020) 043526

¹⁵ Harada, Yoo, Nakao & Jhingan, Phys. Rev. D 96 (2017) 083517

Physical and Theoretical Implications

Relevance to theoretical interests

- No-hair conjecture has extensions and exceptions
 - Hui-Nicolis no-hair conjecture¹²:
 g_{rr} diverges at horizon, therefore finite $J^\mu J_\mu \Rightarrow J^r = 0$ everywhere,
 $J^r = \phi' f(\phi', g, g') \Rightarrow \phi' = 0$
 - Lee-Weinberg charged black hole with scalar hair¹³
- Dyonic (and/or) regular black hole constructions are sought-after
 - Bronnikov condition¹⁴:
“For linear electrodynamics weakly coupled to gravity, regular electrically charged objects cannot obey the Maxwell weak field limit in its core region”

¹² Hui & Nicolis, Phys. Rev. Lett. 110 (2013) 241104

¹³ Lee & Weinberg, Phys. Rev. D 44 (1991) 3159

¹⁴ Bronnikov, Phys. Rev. D 63 (2001) 044005

Evolution and Late-time Properties

Cauchy instability¹⁹:

- causal structure of spacetime within a Cauchy horizon renders the interior unphysical
- observers at the Cauchy horizon should encounter infinitely blueshifted radiation influx
- induces an exponentially growing mass function at the Cauchy horizon

Extremal final states proposed as a solution to various horizon instability

- an object such as ours could be expected to evolve towards extremality

Also commonly necessary to remove ghost and Laplacian instabilities²⁰

¹⁹ Penrose, in Battelle Rencontres (1968); Poisson & Israel, Phys. Rev. Lett. 63 (1989) 1663

²⁰ De Felice & Tsujikawa, Living Rev. Relativ. 13 (2010) 3

Summary and Outlook

Starting from the rich phenomenological consequences of axion cosmologies

Role of symmetry breaking and the resulting cosmological defects

We analysed Kogan's monopole bag in gravitational setting

- a deformed axion profile

- potential black hole-like solutions

- possible stable form

Close ties to: regular solutions e.g. gravastars²¹, regular black holes
 other early universe objects e.g. vacuum bubbles from first order phase transitions

Can expand to general case with non-standard couplings (non-linear electrodynamics)

Consideration of UV effects and more realistic core model

²¹ Mazur & Mottola, Proc. Natl. Acad. Sci. USA 101 (2004) 9545

Appendices

① Classification of Topological Defects

② More on the Strong CP Problem

③ Cauchy Instability

④ Callan-Rubakov Effect

⑤ Instantons and More on PQ

Classification of Topological Defects



Topological defects are classified by their homotopy class²²

For a field theory described by a continuous symmetry group G spontaneously broken to subgroup $H \subset G$

Space of all accessible vacua of the theory (the vacuum manifold) is $G/H = M$

If the vacuum manifold has nontrivial homotopy group $\pi_n(M)$ – equivalence classes of maps of S^n into M (equivalent $\Rightarrow$ can be smoothly deformed into each other)

$\pi_0(M) \neq 1$	Domain wall
$\pi_1(M) \neq 1$	Cosmic string
$\pi_2(M) \neq 1$	Monopole
$\pi_3(M) \neq 1$	Texture

²² Kibble, J. Phys. A 9 (1976) 1387

More on the Strong CP Problem

QCD has the term $\mathcal{L}_{\bar{\vartheta}} \sim \bar{\vartheta} G \tilde{G}$, $\tilde{G}^a_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} G^{a\rho\sigma}$

$G \tilde{G}$ is CP odd so there is an effective angular contribution $\arg \det M_q$ from quark masses, which is at odds with small $\bar{\vartheta}$ required by experiments

PQ introduces anomalous current under QCD $\partial_\mu J_{PQ}^\mu \sim G \tilde{G}$ so that the axion couples to gluons as

$$\mathcal{L} \supset \left(\bar{\vartheta} + \frac{a}{f_a} \right) G \tilde{G}$$

Non-perturbative QCD contains topologically distinct vacuum sectors connected by tunnelling configurations (instantons) effects

Summing over these give the periodic potential

$$V(a) = \chi \left(1 - \cos \left(\bar{\vartheta} + \frac{a}{f_a} \right) \right)$$

→ Topological susceptibility

which minimises at $\left\langle \bar{\vartheta} + \frac{a}{f_a} \right\rangle = 0$ such that the effective value is 0, dynamically solving the strong CP problem

Cauchy Instability

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Also known as mass “inflation instability”

Separating (time dependent) metric such that spherical symmetry manifest:

$$ds^2 = g_{ab}dx^a dx^b + r^2 d\Omega^2$$

such that, for our setup

$$g_{ab}r^{,a}r^{,b} = C = 1 - \frac{2M(r)}{r} \quad \kappa \equiv -\frac{\sqrt{N}C'}{2}$$

→ Misner-Sharpe mass
→ Surface gravity

Can then write in form $ds^2 = NCdudv + r^2 d\Omega^2$ using Eddington-Finkelstein coordinates u and v

which can then be Kruskalised as $ds^2 = \frac{1}{\kappa^2} dUdV + r^2 d\Omega^2$ using $U = -e^{-\kappa u}$, $V = -e^{-\kappa v}$ so that $NC \simeq -2UV$

In the near-horizon regions, the matter part of the energy-momentum tensor consists of crossflowing null fluxes:

$$T_{ab} = \rho_{in} \partial_a V \partial_b V + \rho_{out} \partial_a U \partial_b U$$

We can then define luminosities related to mass

$$\rho_{out} = \frac{L_{out}(U)}{4\pi r^2} \quad \rho_{in} = \frac{L_{in}(V)}{4\pi r^2} \quad L_{in}(V) = \left(\frac{dv}{dV}\right)^2 \frac{d\bar{M}_{in}(v)}{dv} \quad \Rightarrow L_{in}(V) \sim e^{\kappa v}$$

Callan-Rubakov Effect



An SU(2) 't Hooft-Polyakov monopole could induce protons decay at rates comparable to that of the strong interaction

Massless fermions interacting with a monopole should violate baryon number conservation

Where $\vec{L}$ is orbital angular momentum, $\vec{s}$ is the spin of the fermion, total angular momentum is

$$\vec{J} = \vec{L} + \vec{s} + \frac{eg\vec{r}}{r}$$

Charge field interaction (final term) contributes half integer spin by Dirac quantisation

GUT monopole has an indefinite fermion number, so a fermion passing through will be changed
Mimic the Witten effect at low energies due to charge deposition on the monopole

Dyons so formed cannot be stable

May be formulated in a curved background

Modified profile for the fermion condensate exterior to the monopole

Instantons and More on PQ



QCD contains

$$\mathcal{L}_\vartheta = \vartheta \frac{g_s^2}{32\pi^2} G_a^{\mu\nu} \tilde{G}_{a\mu\nu}$$

so each instanton contributes $e^{i\vartheta Q}$, $Q = \frac{g_s^2}{32\pi^2} \int d^4x G_a^{\mu\nu} \tilde{G}_{a\mu\nu}$
and the partition function looks like

$$Z(\vartheta) = \sum_Q e^{-S_I} e^{i\vartheta Q}$$

→ Euclidean instanton action

Summing over all independent instantons, the maths gives

$$Z(\vartheta) \sim \exp(V_4 \cos \vartheta)$$

Then using the standard result

$$E(\vartheta) = -\frac{1}{V_4} \ln Z$$

→ Euclidean 4-volume

we recover the potential of the form

$$V(\vartheta) \sim 1 - \cos \vartheta$$